15-442/15-642: Machine Learning Systems

Data Layouts

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Outline

Tiled & Thread Layout

Distributed Layout

Swizzle Layout

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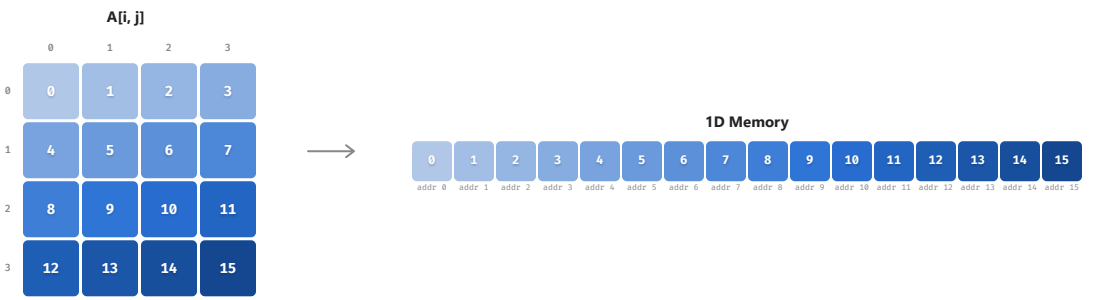
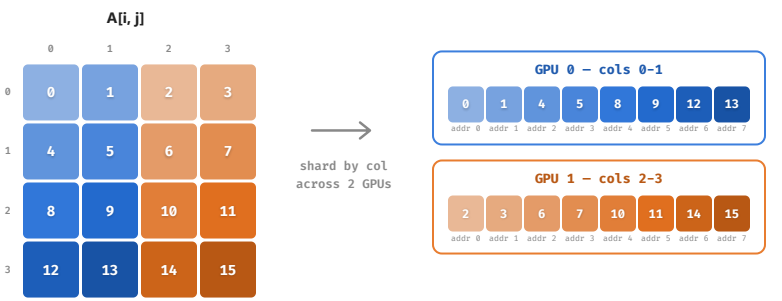
Swizzle Layout

What is Data Layout

Mapping from logical indices to the phyiscal location of the data



or different devices...

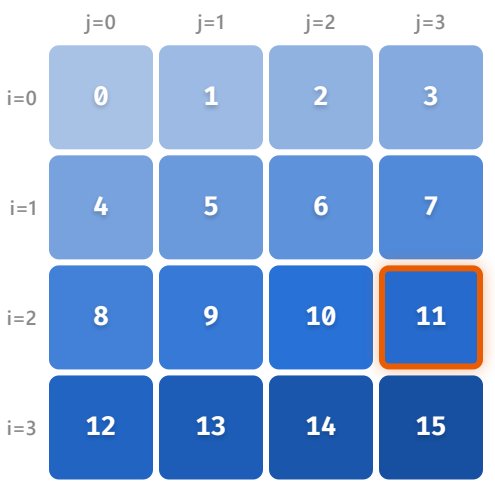


Shape–Stride Model

We will use this notation:

$$S[(4,\ 4)\ :\ (4,\ 1)]$$

shape stride



Address computation

$$addr(i, j) = i \times 4 + j \times 1$$

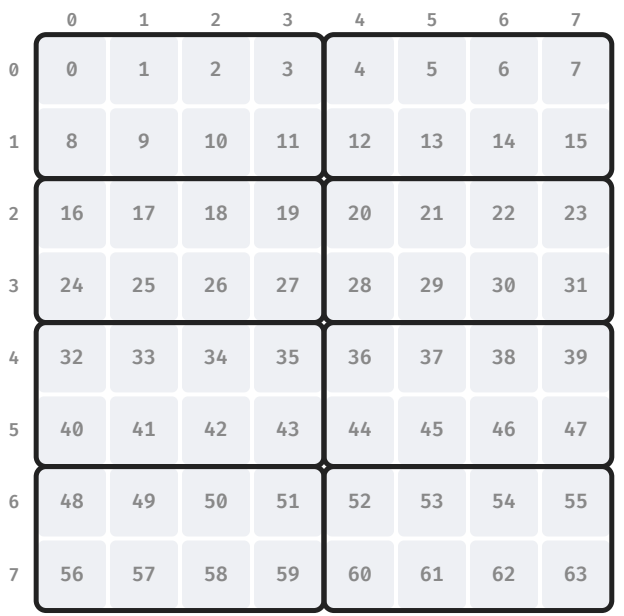
Example: $addr(2, 3) = 2 \times 4 + 3 \times 1 = 11$

This notation can be viewed as simplfied version of CuTe, with difference that we adopt row-major ordering.

Tile Layout

Representing tiling

Layout: S[(4, 2, 2, 4) : (16, 4, 8, 1)]



Address computation

Original shape (8 , 8)

↓ reshape

Tiled shape (4 , 2 , 2 , 4)

Mapped indices (i//2 , i%2 , j//4 , j%4)

addr = (i//2)×16+(i%2)×4+(j//4)×8+(j%4)×1

Example: addr(2, 3) = 1×16 + 0×4 + 0×8 + 3×1 = 19

Tile Layout: Interactive Example

S[(4, 2, 2, 4) : (16, 4, 8, 1)]

Logical view vs Physical memory | click cell to see mapping

Logical Matrix

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	1	2	3	4	5	6	7
r1	8	9	10	11	12	13	14	15
r2	16	17	18	19	20	21	22	23
r3	24	25	26	27	28	29	30	31
r4	32	33	34	35	36	37	38	39
r5	40	41	42	43	44	45	46	47
r6	48	49	50	51	52	53	54	55
r7	56	57	58	59	60	61	62	63

Physical Memory (tiled 2×4)

	+0	+1	+2	+3	+4	+5	+6	+7
0	0	1	2	3	8	9	10	11
8	4	5	6	7	12	13	14	15
16	16	17	18	19	24	25	26	27
24	20	21	22	23	28	29	30	31
32	32	33	34	35	40	41	42	43
40	36	37	38	39	44	45	46	47
48	48	49	50	51	56	57	58	59
56	52	53	54	55	60	61	62	63

Click a cell to see how logical element maps to physical memory

Named Axes in Strides

We need to represent different hardware axese:

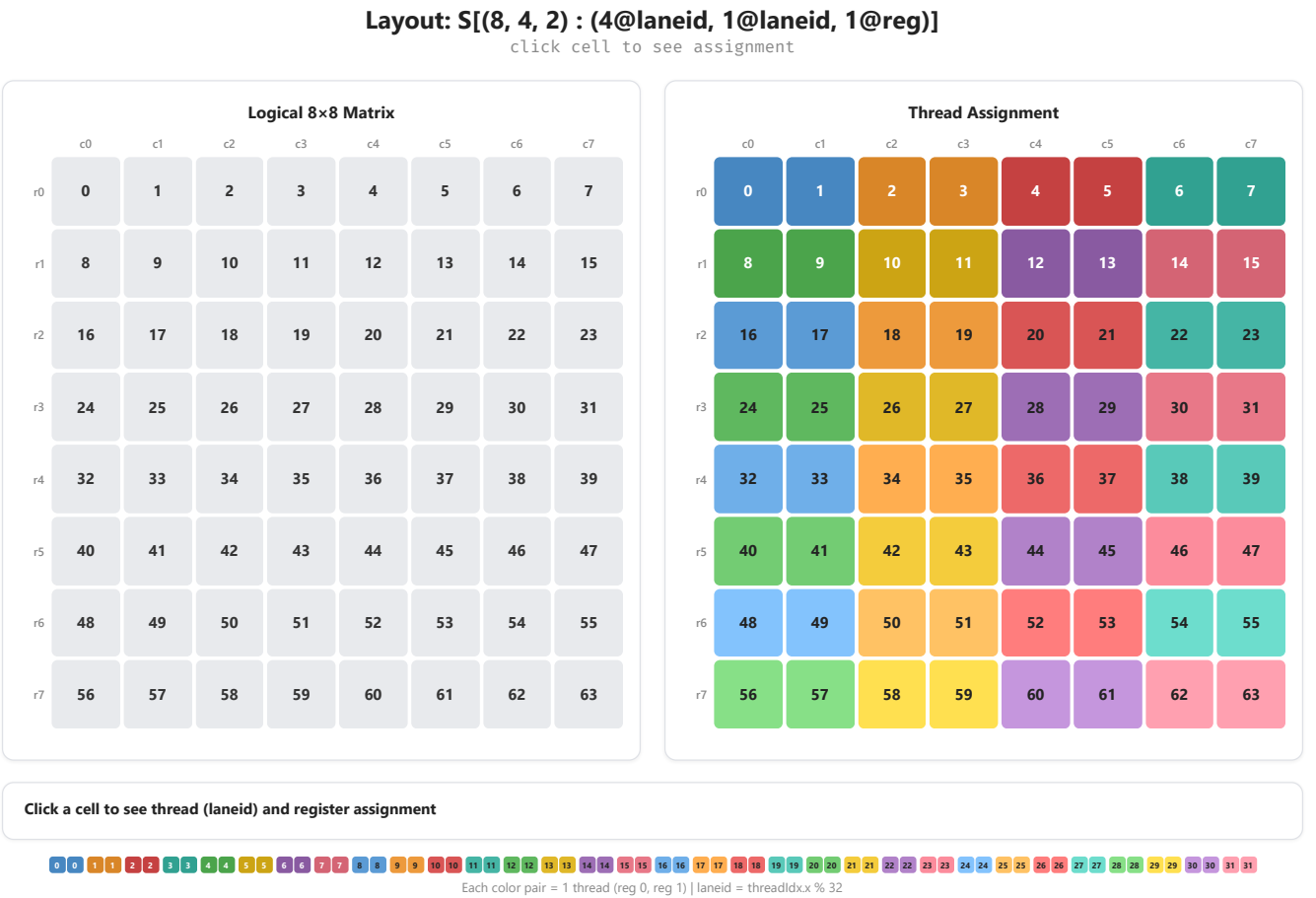
- Lane and col dimension in tensor memory
- Thread indcies

It is useful to have named axes, we use **@m** for normal meory

- `Row-major 8×16: S[(8, 16) : (16@m, 1@m)]`

The `@axis` tag can also be other things like `@warpid`, or `@tmemcol`

Named Axes Example: Thread + Register



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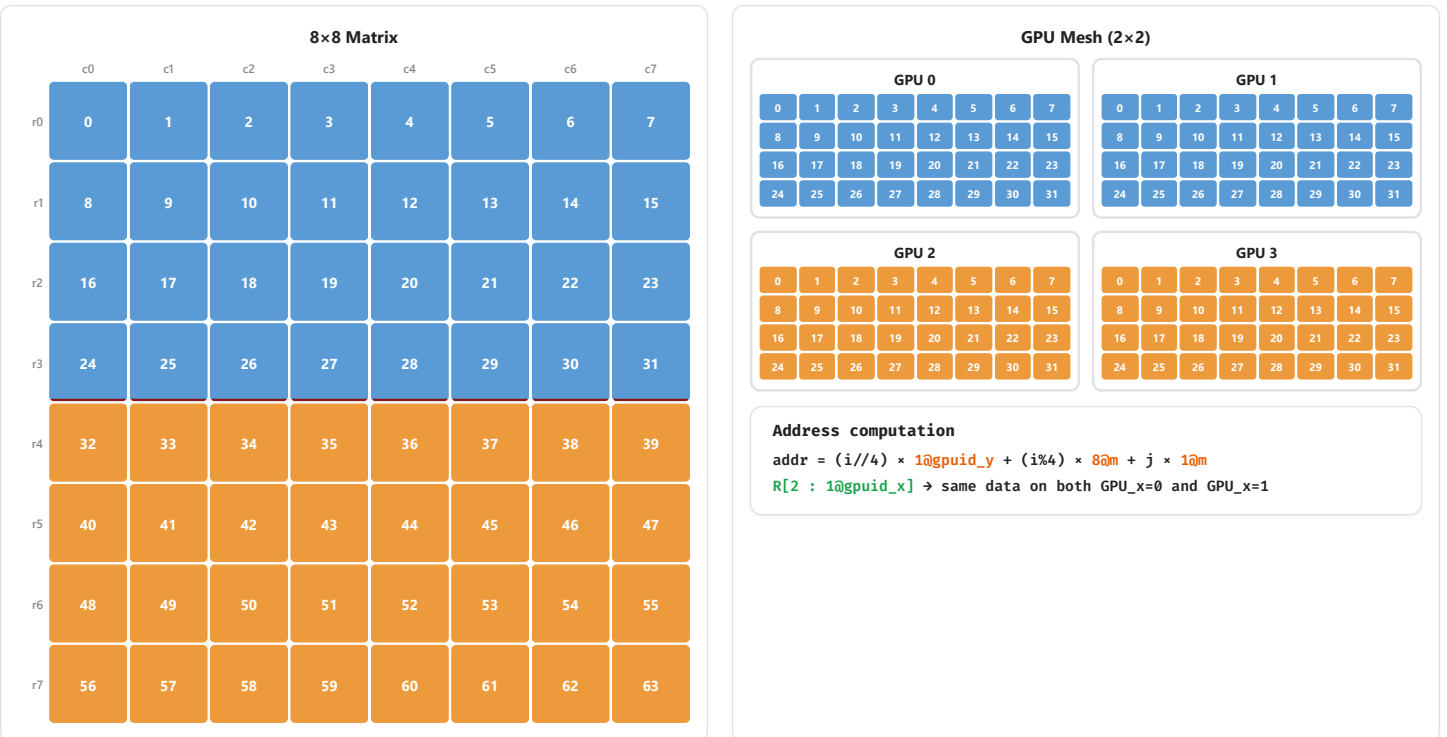
Distributed Layout

Swizzle Layout

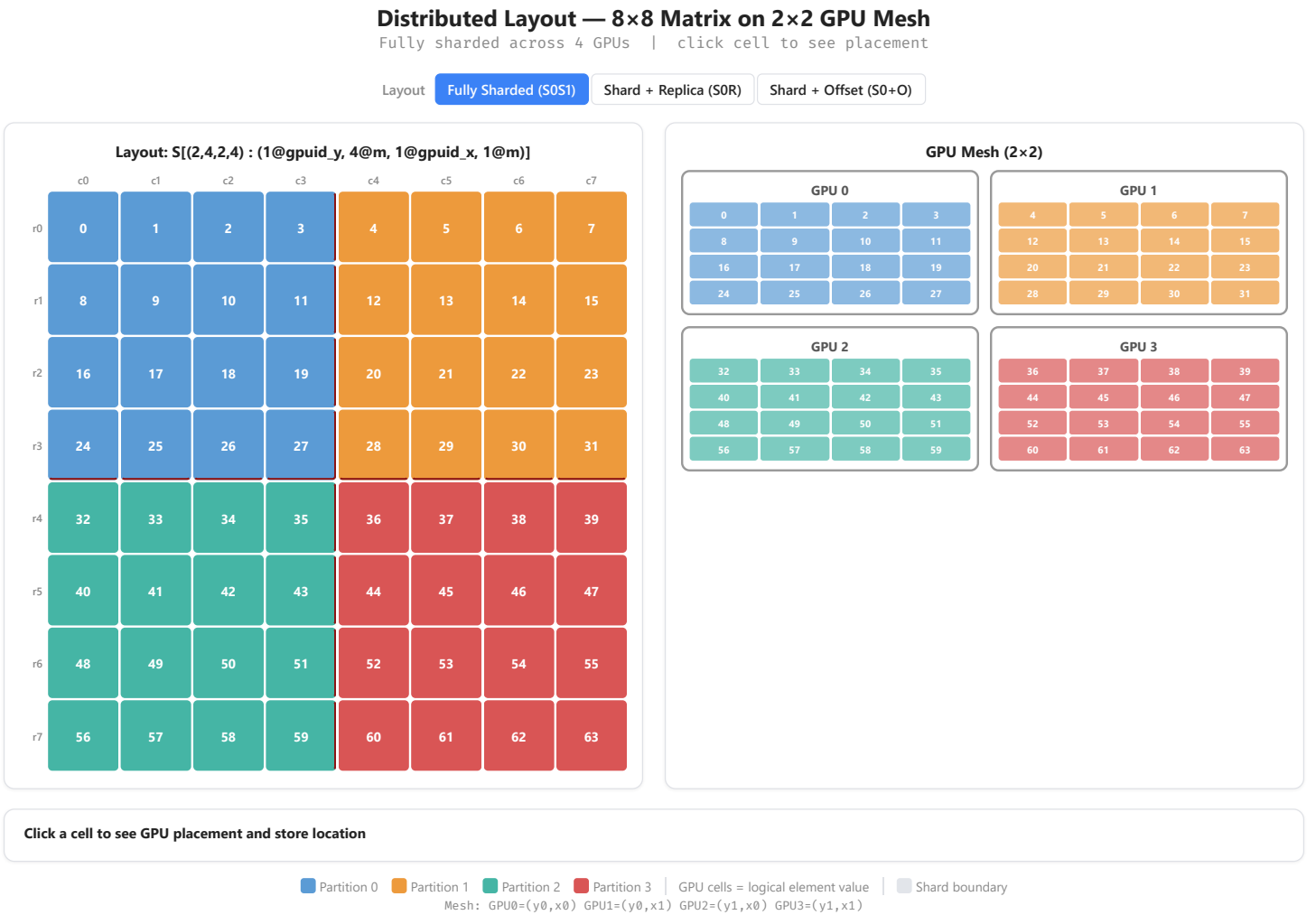
Distributed Axes: @gpuid_y, @gpuid_x

Representing broadcast / replication

Layout: S[(2, 4, 8) : (1@gpuid_y, 8@m, 1@m)] + R[2 : 1@gpuid_x]



Distributed Axes: Interactive



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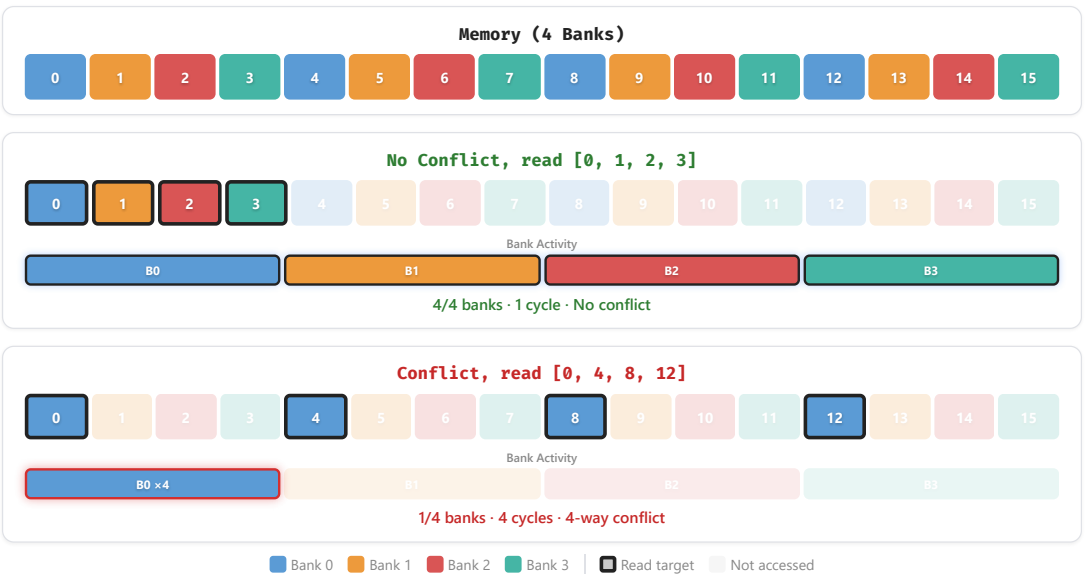
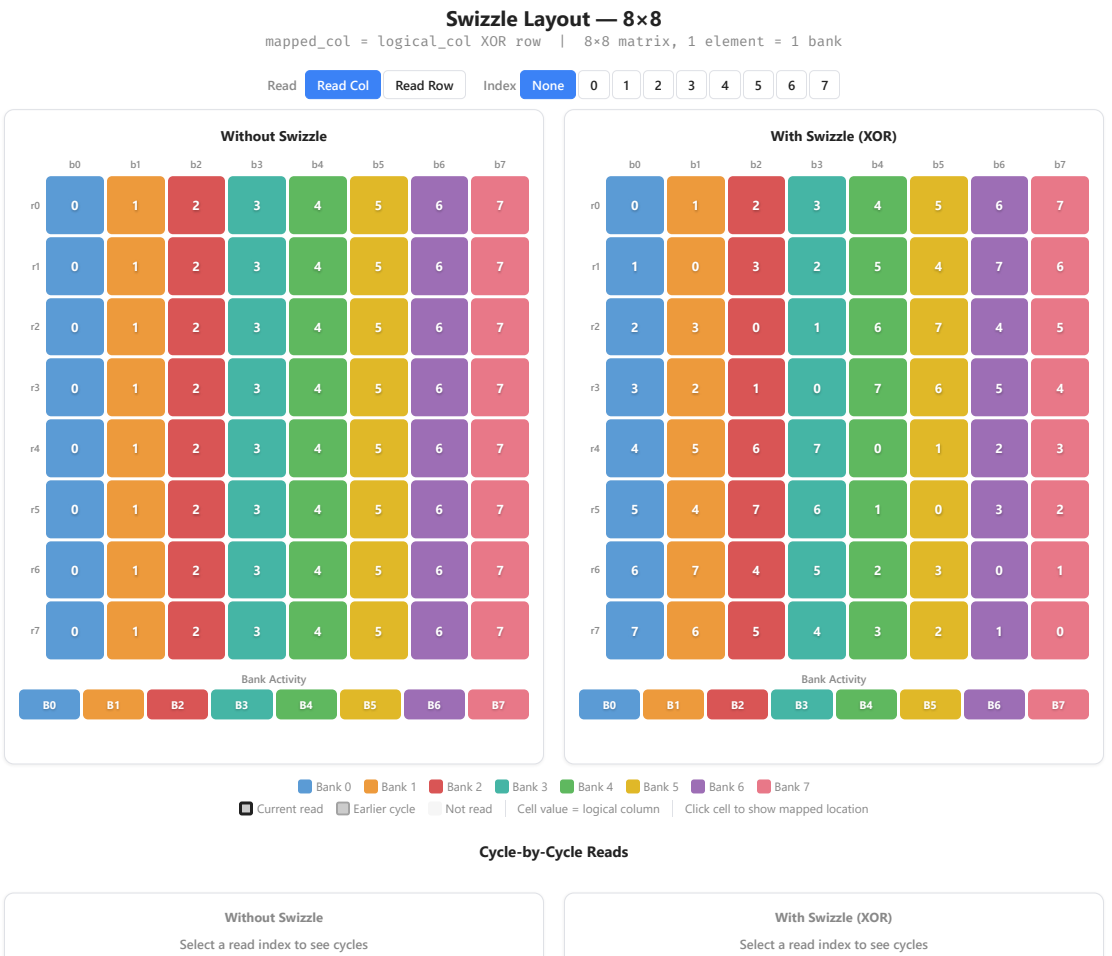
Background: Memory Banks

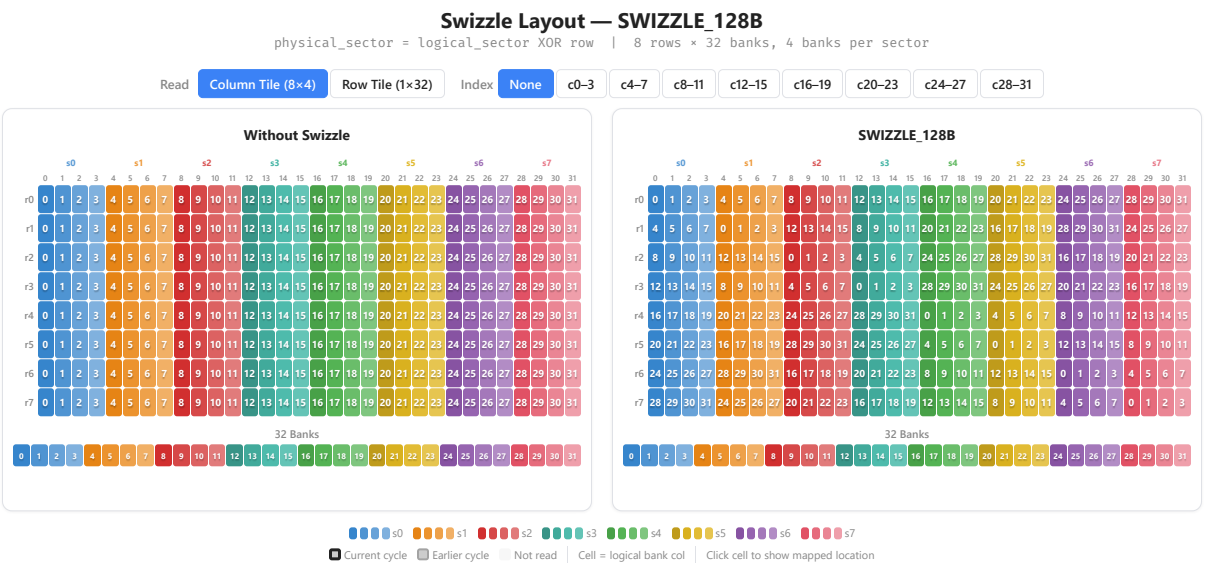
Different memory banks can be read out in parallel



Simple Swizzle: 8×8 Interactive Example



Real Example: SWIZZLE_128B



1 element = 1 bank (4 bytes). A warp of 32 threads reads 32 elements; bank conflicts serialize into extra cycles.
Column Tile (8×4): 8 rows × 4 cols (1 sector) = 32 reads. With swizzle: **1 cycle**. Without: **8 cycles** (8-way conflicts).
Row Tile (1×32): 1 row × 32 cols = 32 reads. Always **1 cycle** (all 32 banks distinct within a row).

Swizzle as a Memory Optimization

Swizzle is an implicit remapping the hardware applies — just enable it

- Pass pre-swizzled addresses and use a consistent swizzle mode (e.g. SWIZZLE_128B) across ops
- Don't worry about swizzle yourself — let hardware handles it.
- Benefit: SWIZZLE_128B → conflict-free access to 8 rows and 8 columns at a time (fp16)

Swizzle is fundamentally a way to enable efficient column access — the idea is not specific to GPU shared memory!